

4.6.2 Triple Integrals in Spherical Coordinates Worksheet

1. Change the following points from rectangular coordinates to spherical coordinates.

a) $(0, -2, 0)$

b) $(-1, 1, -\sqrt{2})$

c) $(1, 0, \sqrt{3})$

d) $(\sqrt{3}, -1, 2\sqrt{3})$

2. True or False: Every point in $\mathbb{R}^3$ can be uniquely represented in spherical coordinates.

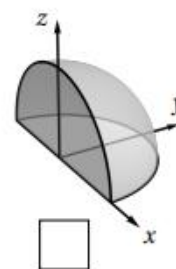
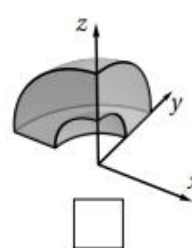
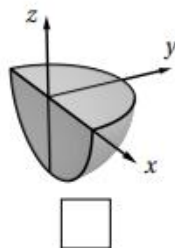
3. Describe in words the surface whose equation is given.

a) $\phi = \frac{\pi}{3}$

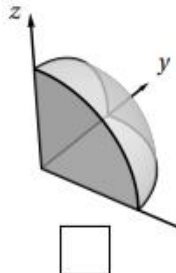
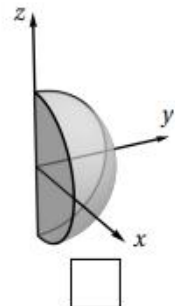
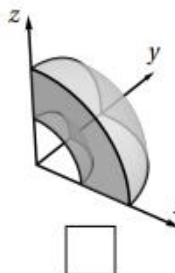
b) $\rho = 3$

4. For each of the given integrals, label the box below the picture of the corresponding region of integration in spherical coordinates.

a) $\int_0^{\pi/2} \int_0^{\pi/2} \int_1^2 \rho^2 \sin(\phi) d\rho d\phi d\theta$

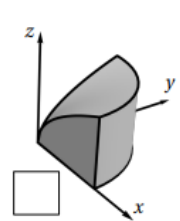
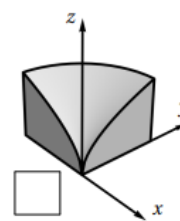
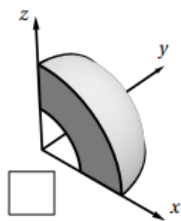


b) $\int_0^{\pi} \int_{\pi/2}^{\pi} \int_0^2 \rho^2 \sin(\phi) d\rho d\phi d\theta$

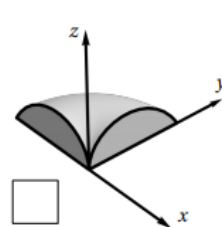
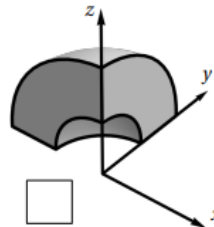
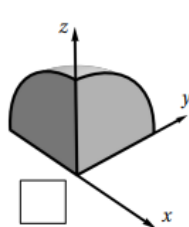


5. For each of the given integrals, label the box below the picture of the corresponding region of integration in spherical coordinates.

a) $\int_1^2 \int_0^{\pi/2} \int_{\pi/2}^{\pi} f(\rho, \phi, \theta) \rho^2 \sin(\phi) d\theta d\phi d\rho$



b) $\int_{\pi/2}^{\pi} \int_0^{\pi/2} \int_0^{\phi} g(\rho, \phi, \theta) \rho^2 \sin(\phi) d\rho d\phi d\theta$



6. The triple integral $\int_0^{\pi/2} \int_0^{\pi/2} \int_0^2 \rho \sin(\phi) d\rho d\theta d\phi$ gives the volume of: (select which answer is correct)

- a) half a sphere of radius 2.
- b) One eighth of a sphere of radius 2.
- c) One quarter of a sphere of radius 2.

7. Sketch the solid whose volume is given by the integral and then evaluate the integral.

a)
$$\int_0^{\pi/6} \int_0^{\pi/2} \int_0^3 \rho^2 \sin(\phi) d\rho d\phi d\theta$$

b)
$$\int_0^{2\pi} \int_{\pi/2}^{\pi} \int_1^2 \rho^2 \sin(\phi) d\rho d\phi d\theta$$

8. Setup, but do not evaluate $\iiint_E (9 - x^2 - y^2) dV$, where E is the solid hemisphere $x^2 + y^2 + z^2 \leq 9, z \geq 0$.

9. Evaluate $\iiint_E (x^2 + y^2) dV$, where E lies between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 9$.

10. Evaluate $\iiint_E y^2 dV$, where E is the solid hemisphere the spheres $x^2 + y^2 + z^2 \leq 16$, $y \geq 0$.

11. Evaluate $\iiint_E x e^{x^2 + y^2 + z^2} dV$, where E is the portion of the unit ball $x^2 + y^2 + z^2 \leq 1$ that lies in the first octant.

12. Evaluate $\iiint_E xyz \, dV$, where E lies between the spheres $\rho = 2$ and $\rho = 4$ and above the cone $\phi = \frac{\pi}{3}$.

13. Find the volume of the part of the ball $\rho \leq 2$ that lies between the cones $\phi = \frac{\pi}{6}$ and $\phi = \frac{\pi}{3}$.

14. Convert each of the following integrals to cylindrical coordinates.

a)
$$\int_0^3 \int_{-\sqrt{9-z^2}}^{\sqrt{9-z^2}} \int_{-\sqrt{9-y^2-z^2}}^{\sqrt{9-y^2-z^2}} x^2 dx dy dz$$

b)
$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{-\sqrt{4-x^2-y^2}}^{\sqrt{4-x^2-y^2}} (x^2 + y^2 + z^2)^{3/2} dz dy dx$$

$$\text{c) } \int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} xy \, dz dy dx$$